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B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

SE 19-206-0405 ELECTRICAL TECHNOLOGY AND SAFETY (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Explain the constructional features and applications of electrical machines.
 CO2: Solve numerical problems related to the operation of electrical machines.
 CO3: Explain the working principle of various electrical safety equipments.
 CO4: Summarise the safety precautions to be taken during installation of plants and equipments.
 Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
 PO – Programme Outcome

PART A(Answer *ALL* questions)

(8 × 3 = 24)

		Marks	BL	CO	PO
I.	(a) Discuss the role and functioning of a current transformer (CT) in an electrical power system.	3	L1	1	3
	(b) Describe the flux control method for varying speed of DC motor.	3	L1	1	1
	(c) Why is it important to consider restriking voltage in circuit breaker design? Discuss the influencing factors also.	3	L2	2	3
	(d) Discuss how these faults occur in power systems. Explain examples of symmetrical faults.	3	L3	2	2
	(e) Discuss the properties, benefits and applications of FRLS insulation.	3	L2	3	3
	(f) Evaluate the function spark gap as a protective device in electrical systems.	3	L2	3	1
	(g) Enumerate some key safety provisions outlined in the Indian Electricity Act and Rules to ensure electrical safety.	3	L1	4	1
	(h) Discuss the concept of energy limitation as it applies to intrinsically safe devices.	3	L1	4	3

PART B

(4 × 12 = 48)

II.	The following data were obtained on a 50 kVA, 2400/120 V transformer. Compute the equivalent circuit parameters referred to high voltage side. OC test: 120 V, 9.65 A, 396 W SC test: 92 V, 20.8 A, 810 W.	12	L4	1	2
OR					
III.	(a) A 20 hp, 230 V, 1150 rpm shunt motor has 4 poles, 4 parallel armature paths and 882 armature conductors. The armature circuit resistance is 0.188 Ω. At rated speed and rated output the armature current is 73 A and field current is 1.6 A. Calculate the flux.	6	L3	1	2
	(b) Discuss the factors that influence the generated electromotive force (EMF) in a DC generator, including the number of turns in the armature coil, magnetic field strength, speed of rotation and flux distribution. Provide examples of how variations in these factors impact the output voltage of a DC generator and discuss the practical applications of the EMF equation.	6	L2	1	2

		Marks	BL	CO	PO
IV.	(a) A balanced 3 phase, 60 Hz voltage is applied to a 3 phase, 4 pole induction motor. The slip of the machine is found to be 0.05, when motor delivers rated output horse power. Determine the following (i) the speed of revolving flux relative to stator structure. (ii) frequency of rotor currents. (iii) the speed of rotor mmf relative to rotor structure. (iv) the speed of rotor mmf relative to stator structure.	6	L3	2	3
	(b) Illustrate and explain the relationship between slip and torque in an induction motor with variation of slip with torque across different operating conditions and load levels. Also explain the significance of slip in the performance and efficiency of induction motors.	6	L4	2	2
OR					
V.	(a) Discuss the principles of operation, types, installation considerations and advantages of ELCBs in protecting against electric shock and electrical fires. Provide examples of scenarios where ELCBs are essential for ensuring safe electrical installations and compliance with safety standards.	6	L2	2	3
	(b) Describe with neat diagram the construction, working principle, applications, advantages and limitations of oil circuit breakers in electrical engineering.	6	L2	2	3
VI.	(a) Explain the principle and application of an arc suppression coil grounding system in electrical installations. How does it enhance safety and protect equipment from electrical faults? Discuss with a diagram illustrating the configuration and components grounding setup.	6	L3	3	1
	(b) Describe the concept of reactance grounding in electrical power systems with a diagram. Discuss advantages and how it differs from other grounding methods. List the examples of where reactance grounding is typically used and explain the considerations in its design and implementation for ensuring safety and reliability in electrical installations.	6	L2	3	1
OR					
VII.	(a) Describe the role and operation of distance relays in electrical transmission systems with a one line diagram. How do distance relays contribute to the protection of power lines from faults?	6	L2	3	3
	(b) What are the main components of a protective relay system used in electrical substations? Provide a labeled diagram illustrating its key elements and their functions in ensuring grid reliability and safety.	6	L2	3	3
VIII.	(a) Discuss the principles and criteria used to classify hazardous areas based on the presence of flammable gases, vapors, liquids, combustible dusts, with typical examples.	6	L2	4	1
	(b) Suggest various methods for preventing following situation: An electrical equipment is placed in a hazardous area, a fault could cause sparks sufficient to ignite the gas and produce an explosion, causing serious damage to people and the environment.	6	L3	4	1
OR					
IX.	(a) What are the hazards associated with static electricity? Discuss its potential risks and methods to control or mitigate static discharge.	6	L1	4	1
	(b) Discuss common relay failure modes and troubleshooting techniques. Explain how preventive maintenance practices can extend relay lifespan and ensure optimal performance in electrical systems.	6	L2	4	3